

Qualidade de Software - 2025/26

Project #1: Human-AI Collaboration in Software Quality

Abstract

Human-Artificial Intelligence (AI) collaboration is becoming a defining characteristic of modern software quality (SQ) activities, as engineers increasingly rely on AI systems to generate, evaluate, and suggest quality-related artifacts such as tests, bug fixes, and reviews. While AI promises efficiency and scalability, it also introduces new socio-technical risks related to trust, over-reliance, and diminished human oversight.

This project investigates how different collaboration models—Human-in-the-Loop (HITL) and Human-on-the-Loop (HOTL)—affect verification behavior, decision quality, and error detection in SQ tasks. Students empirically study how humans supervise, correct, and validate AI outputs, with particular attention to trust calibration and failure modes in human-AI workflows. The assignment aims to produce empirical insights into effective collaboration patterns for AI-assisted software quality assurance (SQA).

Objectives

1. Analyze verification behavior under HITL and HOTL conditions
2. Identify over-reliance and under-verification risks
3. Study trust calibration and decision quality

Student Taks

1. Perform a quality task with AI assistance (e.g., test validation, code review)
2. Execute the task under HITL and HOTL conditions
3. Log decisions, interventions, and rationales
4. Answer a structured reflection questionnaire

Research Questions

RQ1: How does verification behavior differ between HITL and HOTL?

RQ2: What types of AI errors are missed under reduced human oversight?

RQ3: How does trust correlate with decision accuracy?

Data Collected

- Acceptance/rejection rates
- Time-to-decision
- Detected vs missed errors
- Self-reported trust and confidence

Methodology

- Controlled experiment with within-subject comparison
- Quantitative analysis supported by qualitative reflections

Expected Deliverables

- Task artifacts
- Decision and verification logs
- Reflection questionnaire

Assessment Rubric

- Technical / Conceptual Correctness (25%)
Minimum: Correct application of concepts.
Excellence: Nuanced and precise application with justification.
- Empirical Rigor (25%)
Minimum: Data collected and analyzed consistently.
Excellence: Rigorous analysis with clear validity considerations.
- Analytical Depth (20%)
Minimum: Basic interpretation of results.
Excellence: Insightful analysis connecting results to theory.
- Documentation & Communication (15%)
Minimum: Clear and structured report.
Excellence: Publication-quality clarity and coherence.
- Reflection & Insight (15%)
Minimum: Reasonable reflection on findings.
Excellence: Deep reflection on limitations and implications.